09: Evaluating Usability & UX

IN3065/INM355 | 2025–2026
User-Centred System Design

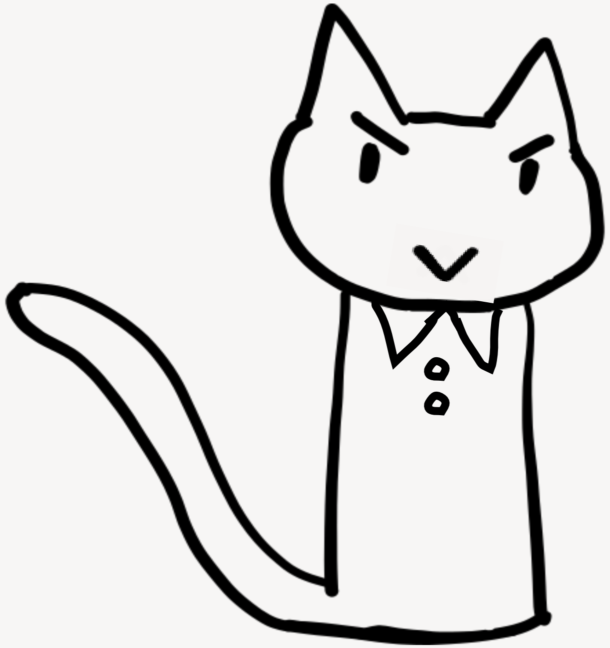
Dr Tracey Booth (she/her)
Centre for HCI Design | Computer Science | SST
tracey.booth.2@citystgeorges.ac.uk

Module Evaluation Questionnaire

Please fill this in ASAP

https://city.surveys.evasysplus.co.uk

Closes Sunday 12th April



Session 9: Learning objectives

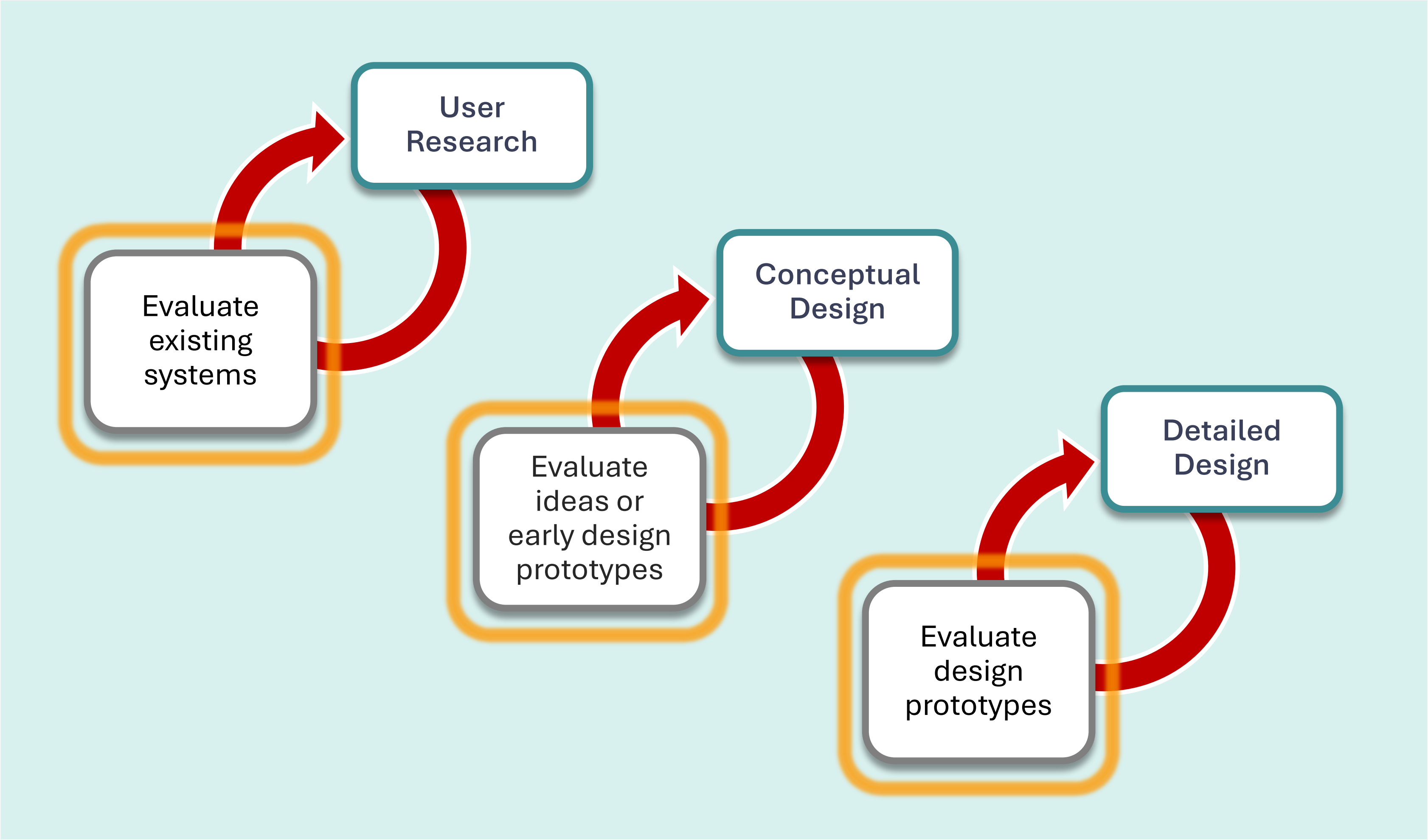
By completing this session, including any practical activity and assigned readings, you should be able to:

- Describe the foundational concepts of usability evaluation.
- Use a structured framework to plan usability testing.
- Analyse data from usability testing sessions
- Design or choose a questionnaire for use in usability evaluation

Evaluation (quick recap)

(Recap)

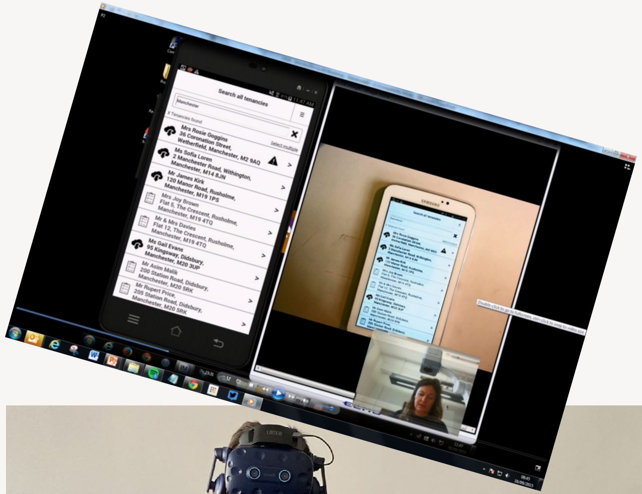
Evaluation in a user-centred design process

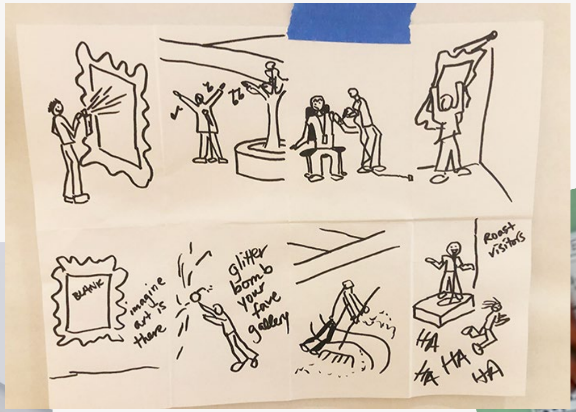


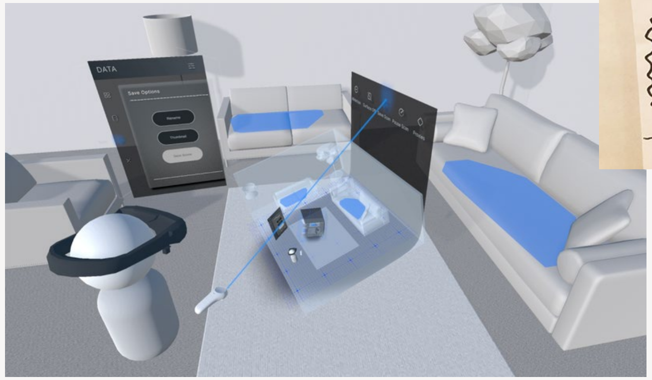
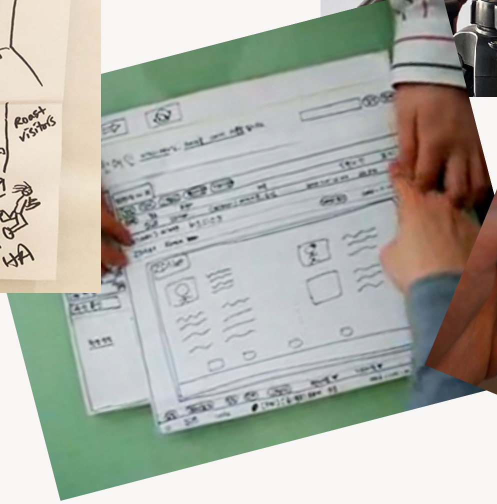
(Recap)

What can we evaluate?

- Sketches
- Paper prototypes
- Wireframes
- Clickable digital prototypes
- Physical prototypes
- Existing systems
- Competitor systems




(Recap)

For what purposes do we evaluate?

- To get early feedback on design concepts.
- To decide between design options.
- To check that users can use the product and whether they like it.
- To identify and fix usability problems in designs *before* production, or release.
- To identify and fix usability problems in our existing products.
- To assess competitor systems.
- To focus on real problems—not our intuition about what is good or bad.

Evaluation 2

Usability & UX Evaluation

Evaluating usability & UX

- As we know, usability and user experience are both important
- Usability evaluation is how we find out about the usability and user experience of interactive systems.
- This is a general term to describe the many different approaches to finding out about usability and user experience.

Methods for evaluating usability & UX

- Can be broadly split into:
  - Evaluation methods that involve users
  - Evaluation methods that don’t involve users
- Different methods can complement each other, and be used at different stages.

Evaluation methods with users

Usually involves users (or representative users) using your design/system

- Usability testing
  Task-based; lab or remote; moderated or unmoderated; often think-aloud
- Query-based techniques
  Questionnaires/Surveys, Interviews (pre/post-task)
- Experiments / A/B testing
  Comparing variants to measure impact
- Analytics / usage data *(if a live system exists)*
  Real user behaviour at scale (e.g. drop-off, time-on-task, click-rate etc)
- Physiological measures *(less common)*
  e.g. eye tracking

Evaluation methods without users

Expert review (inspection methods: Involves experts *inspecting* the design/interface/system

- Heuristic evaluation
  Checking the interface against known usability principles
- Cognitive walkthrough
  Stepping through tasks as if you're the user
- Guidelines / standards review (e.g. WCAG)
  Checking against things like accessibility rules
- Consistency inspection
  Making sure the interface behaves consistently
- UX audit
  A broader expert review of the overall experience

Formative vs summative evaluation

Formative evaluation

- Concerned with improving designs/products.
- Often focused on identifying usability problems; often qualitative data
- Happens early and throughout the design process.
- Results are fed back into the design process.

Summative evaluation

- Concerned with assessing performance/success of a design.
- Often involves measurement (e.g. usability/UX metrics), so usually quantitative data.
- Typically conducted at the end of the design process.
- Results used to judge/compare the design—how well does it perform?
  - Useful for benchmarking and QA.

What does usability evaluation produce?

- Usability evaluation is most often used to find usability problems
  - These are then fixed in the next iteration
  - *Qualitative* data

- Usability evaluation is sometimes used to measure usability & UX, producing metrics
  - To decide whether the usability and UX are good enough
  - *Quantitative* data

Metrics:
Measuring usability and user experience:

Recall the usability goals and aspect of user experience we discussed in lecture 1. We can define and measure metrics such as:

– Effectiveness (success): how often users succeed at tasks.

– Ease of use (efficiency): resources for user to perform a task

– User experience and satisfaction: whether users enjoy using the system, feel successful, in control, competent, etc.

– Ease of learning (learnability): the time and effort required for a user to meet a specified level of performance.

Metrics: Examples

- Effectiveness
  “90% of users find the information they need on Moodle”
- Efficiency
  “On average, it takes a user 2 mins 28 seconds to upload a coursework to Moodle”
- Satisfaction
  “When asked to rate the experience of using Moodle on a scale from 1- 5, where 1 is very unpleasant and 5 is very pleasant, the average rating was 2.8”

Metrics & ‘the user’

Who will use the product?

- Primary users
- Secondary audiences?

What types of metrics might be appropriate for the context of use, i.e. the specific user, their goals and tasks, resources, and the environment in which interaction will take place?

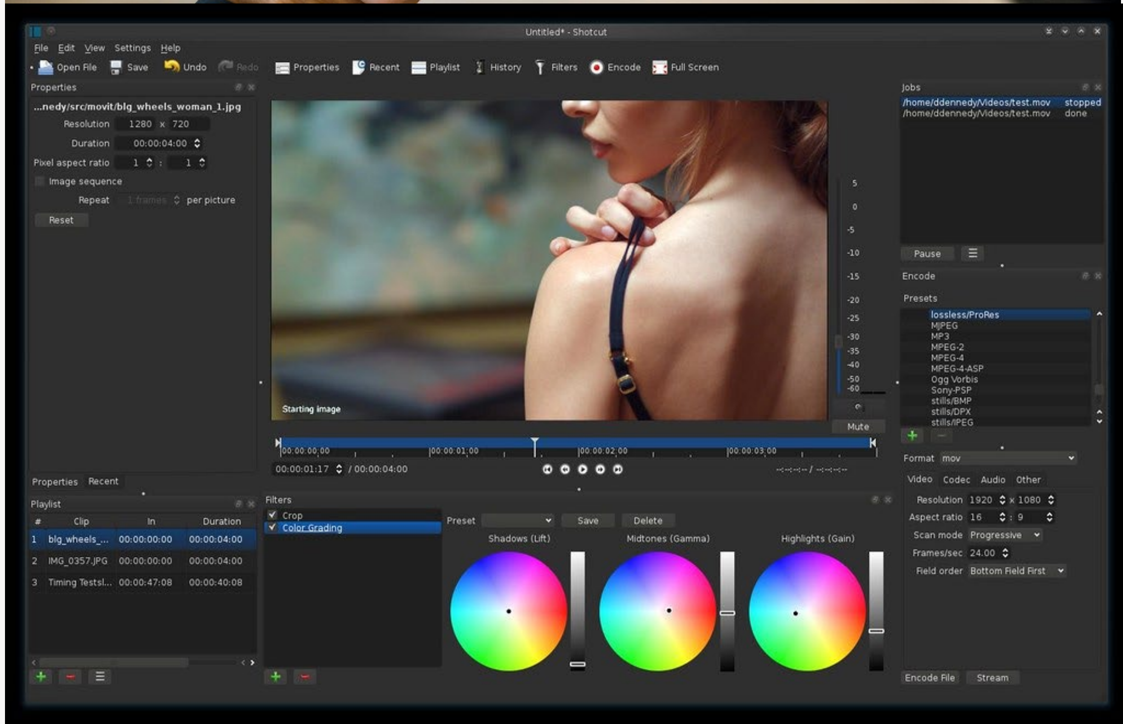
Measuring UX?

Which UX measures would be appropriate for the following, and why?

- A photo editing app?
- A website that allows people to buy and sell vintage clothing?
- A banking app?

https://www.pexels.com/photo/digital-wallet-app-on-smartphone-6406691

https://www.flickr.com/photos/xmodulo/12867967295

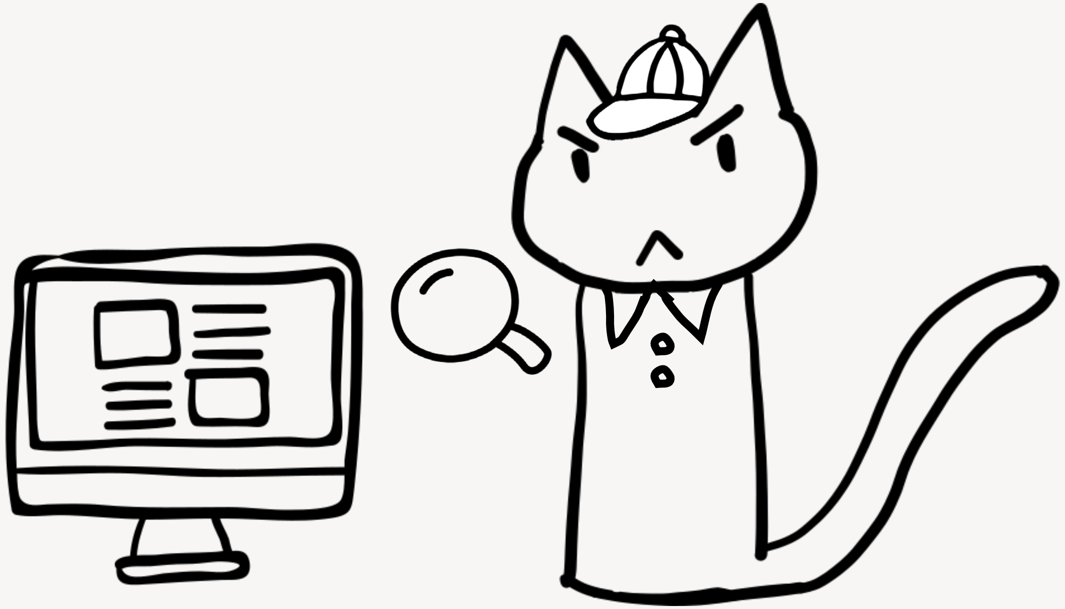


Usability problems: What are they?

- Any aspect of the user interface that causes the system to have reduced usability for the user, and where *changing* it would *improve* usability.
- Where a user has some (even slight) difficulty with an aspect of the system to achieve their goals in typical use.
- Aspects that that make the system (for the user): Less pleasant; Less efficient; More onerous or impossible; More demanding of their attention; Less easy to figure out how to use.
- Identified during usability testing with users, but also predicted by experts using expert review techniques.
- Not the same as *bugs*.

Why find out about usability problems?

- To improve systems by fixing usability problems, as part of iterative design.
- To understand metrics.
- To compare the usability of different products.



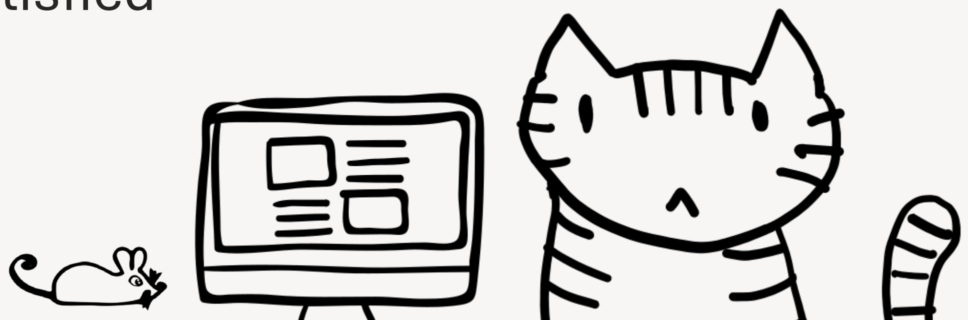
What do usability problems look like?

- *"The user was unable to read the captcha, and therefore could not progress to searching for a ticket"*

- *"The user spent some time before approaching the castle door – did not know how to enter the castle. Eventually, a tip was provided by the facilitator on how to open the door."*

Signs that there are usability problems

- Often, when asked, users will say that there are no problems…
- Hints that there are problems:
  - A user takes a long time to complete what they want to do (e.g. figuring out where to click).
  - A user has difficulty understanding how to use the system (e.g. what to do next).
  - A user makes mistakes (e.g. goes down the wrong path)
  - A user cannot do what they want to do (e.g. gives up)
  - There is evidence that the user is not satisfied (e.g. when they sigh or frown).



Evaluation:
Usability Testing

sometimes referred to as ‘user testing’

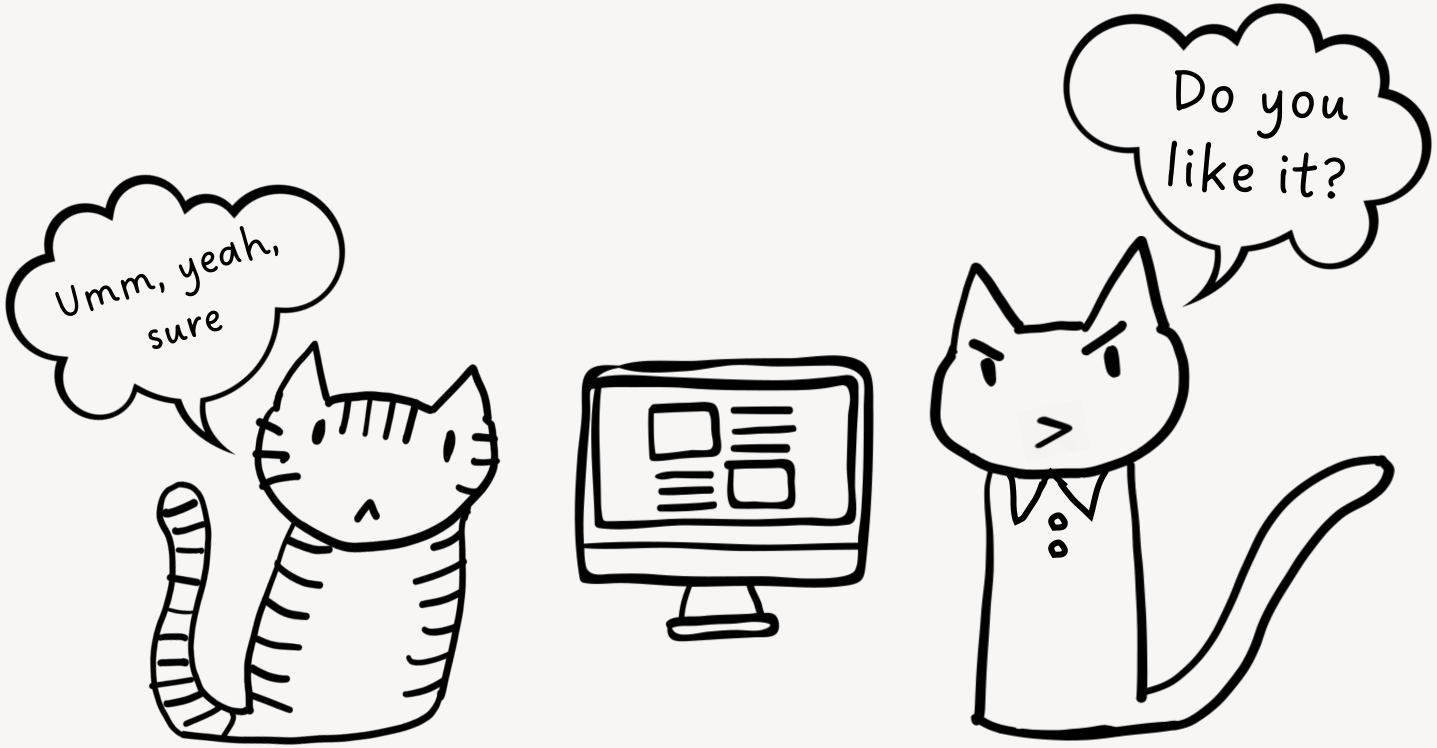
(although we try not to use that term because this isn’t about testing users)

Usability testing: the ‘gold standard’

- Participants undertake realistic tasks
- We observe what they do, and what happens.
- We may also ask them to ‘think aloud’ during the task—what are they thinking?
- Can be conducted
  - In a lab or real-world setting
  - With a prototype or an implemented system
  - Alongside interviews and questionnaires
- Outputs:
  - Evidence-based issues list, including usability problems
  - Can also capture metrics

Usability testing is not...

... showing your product to someone and asking *“what do you think?”*



When can you conduct usability testing?

At the beginning of a project

- Test existing products, or those of competitors, to identify aspects that work well / poorly in order to inform new designs.

During a project

- Test wireframes / prototypes of proposed designs in order to identify / rectify mistakes before building them.

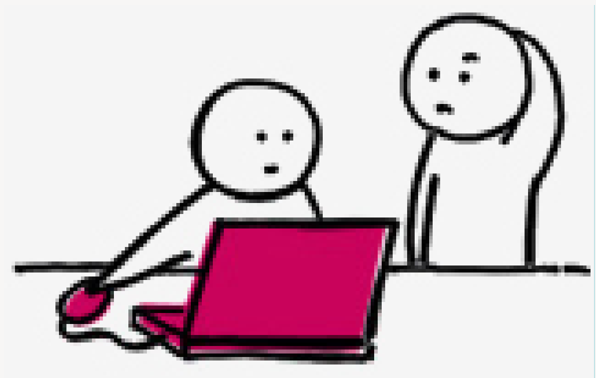
At the end of a project

- Test the implemented designs to ensure implementation matches designs and satisfies project goals.

Types of usability testing

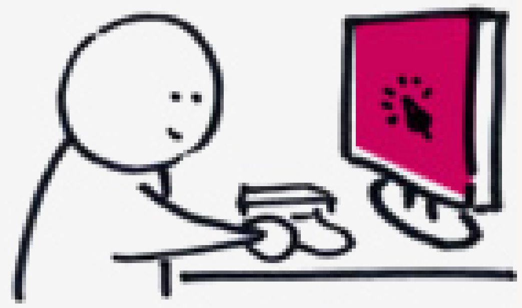
Usability tests can be:

- Face-to-face (aka co-located / in person) or remote
- Moderated or unmoderated



Moderated face-to-face

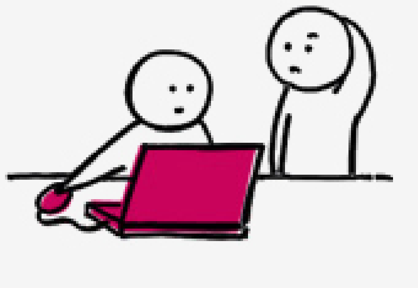
Moderated remote



Unmoderated remote

Images from http://zurb.com/word/unmoderated-user-testing

Traditional (moderated, face-to-face) usability testing



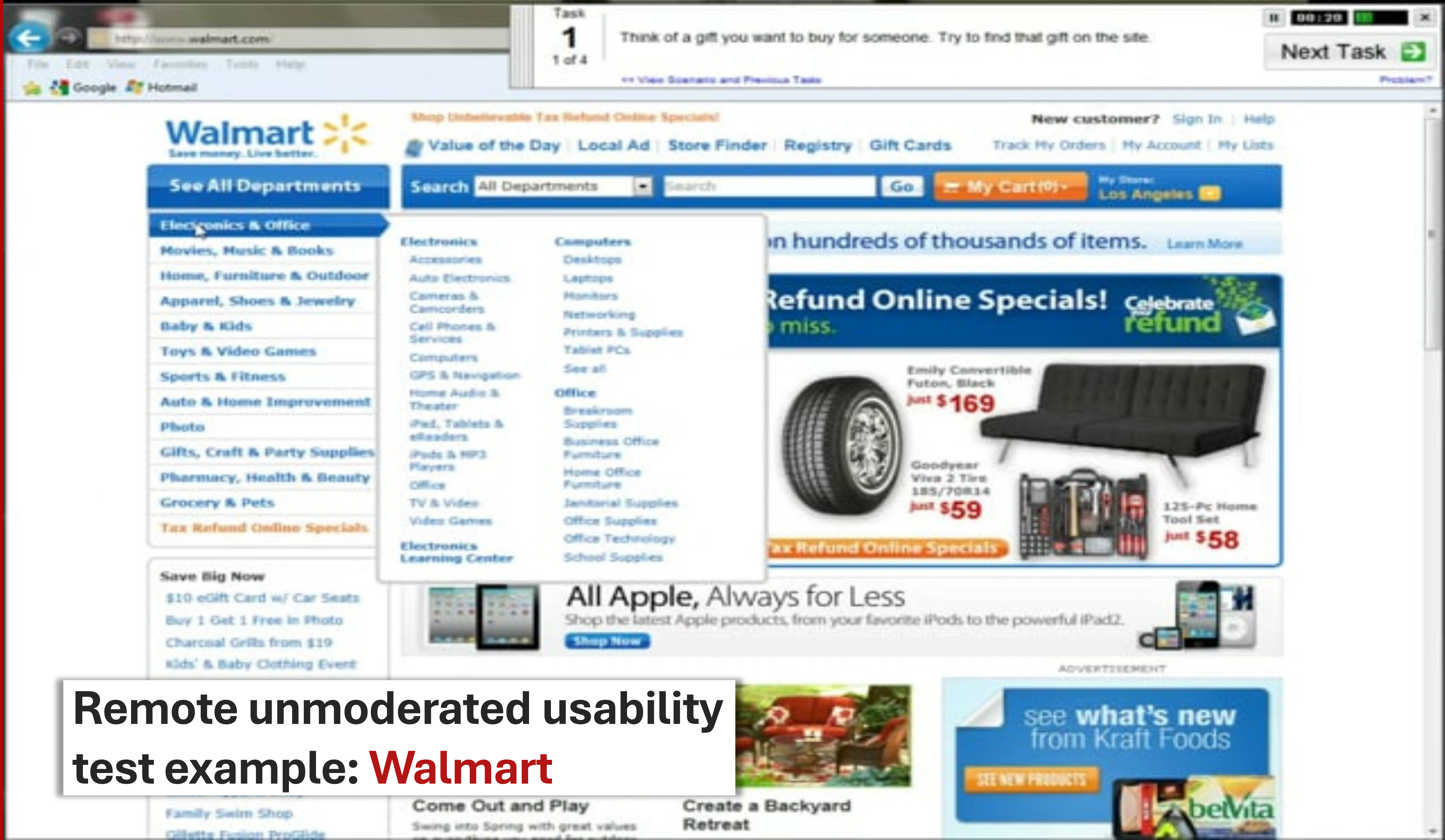
- Participants are recruited, according to a “screener” (covered in lecture 3).
- Often conducted in a controlled environment (e.g. usability lab)
- The moderator observes each participant interacting with the system, to carry out a given set of tasks.
- The participant ‘thinks aloud’ while carrying out the tasks.
- The session is often recorded (video and audio)—both onscreen activity and the participant’s verbal and non-verbal behaviour.
- Moderators note usability problems faced by user.
- User also sometimes complete one or more questionnaires, e.g. to measure satisfaction, and possibly also a debriefing interview.
- Potentially costly to hire, recruit people, pay incentives, etc

Moderated remote usability testing

- Similar to a face-to-face traditional usability test, but the participant (representative user) is online, often at home, and a moderator connected by video conferencing software.
  - e.g. Zoom, MS Teams, Google Meet, etc
- Participant's screen is recorded remotely
- As with a traditional usability test, the moderator can still put the participant at ease, explain tasks, intervene etc.
- But less controlled than traditional usability testing.
  - Beware of internet connection problems and other technical issues, as well as unanticipated interruptions.

Unmoderated remote usability testing

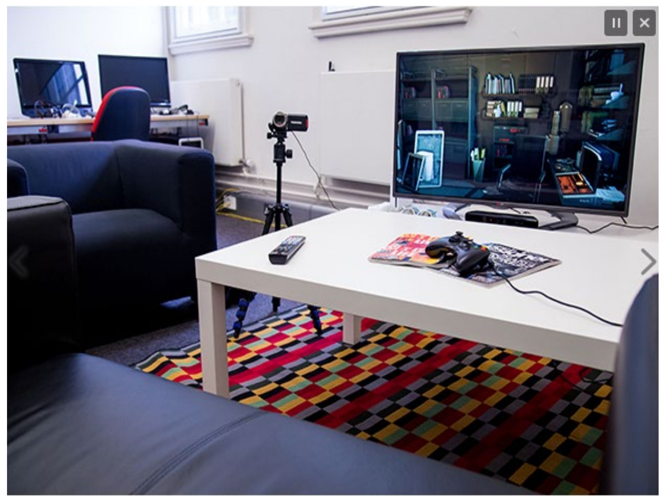
- The participant (representative user) is online, often at home.
- No moderator present, either physically or virtually.
- Software application presents instructions, records participants' actions, and captures questionnaire answers.
- Test is set up in advance, using a particular service.
  - e.g. UserZoom, dscout, UXTweak, Loop11, PlaybookUX, Userlytics...
- Often used to obtain quantitative data (metrics)
- Potentially access to a wider pool of participants.
- BUT no moderator = no observation, intervention, probing or support.
  - Data may be unreliable

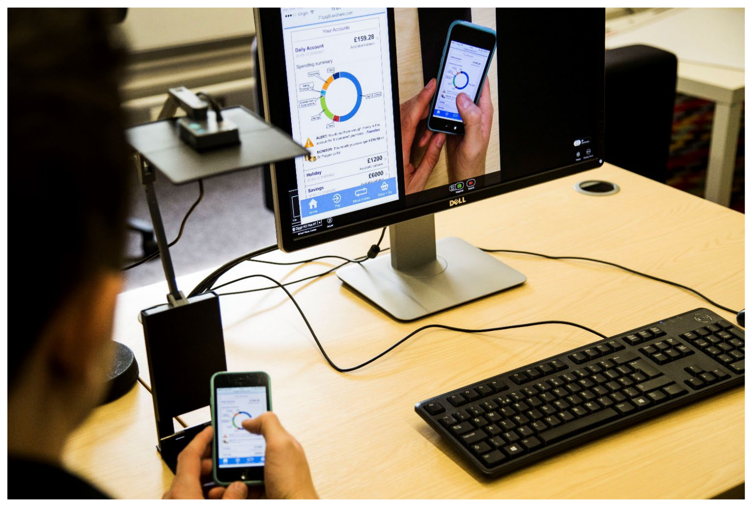


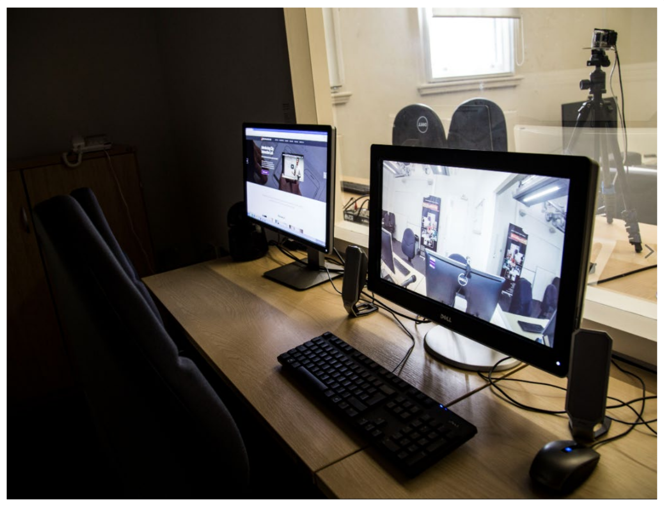
Remote unmoderated usability test example: Walmart

City St George’s Interaction Lab

THE INTERACTION LAB AT CITY ST GEORGE'S







https://www.interaction-lab.co.uk

Usability Testing

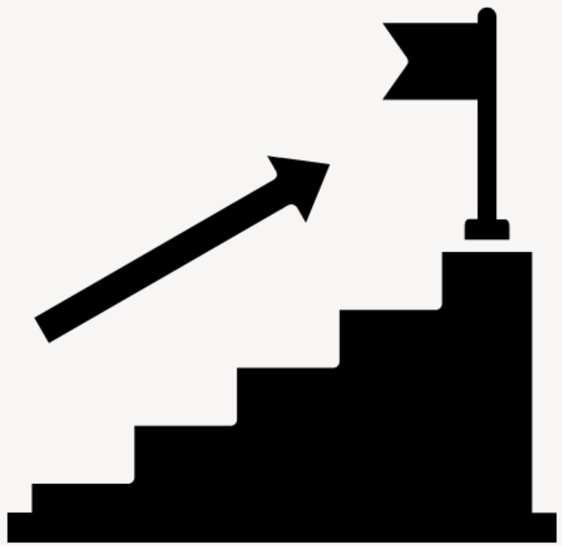
Planning and running a usability test

Planning a usability test

1. What do you want to know? (Goals)
2. What will you test? (Prototype / system)
3. What will you ask people to do, and in what order? (Tasks, Procedure)
4. Who will those people be, and how many of them? (Participants)
5. Where will the testing take place? (Location)
6. How will you conduct the testing session? (Running a session)
7. What will you need? (Equipment & Materials)
8. What ethical issues should you consider? (Ethics)
9. What data will you collect? (Data capture)
10. How will you deal with the data? (Analysis & Reporting)

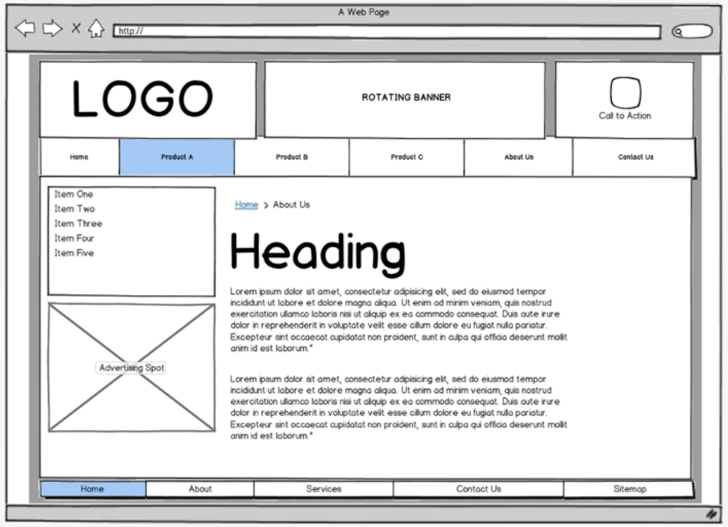
Planning: Goals

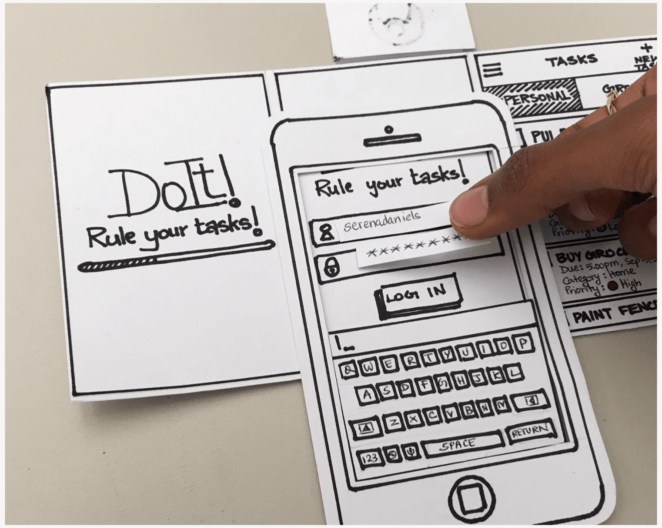
- Be clear about the goals of the evaluation.
- What are you seeking to discover? What findings will you deliver?
  - Usability problems?
  - Metrics? (only if relevant)
  - Answers to research questions?

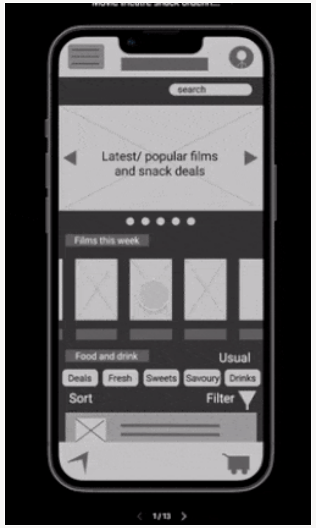


Planning: What to evaluate

- This can be a working system
- But it can also be a wireframe, or a paper prototype, or a clickable digital prototype







Planning: Tasks

- Create tasks for participants to undertake.
  - To make their interaction realistic.
  - To focus them on specific issues / areas of concern.
- Present these tasks as realistic scenarios that are meaningful to users.
- Specificity is important
  - Don’t tell them *exactly* what to do, but don’t leave it too vague.
  - Ask the user to do the task—don’t ask *“How would you…?”*
- Make sure that tasks will be clearly understood.
  - Phrase tasks in the language of users. Watch out for ambiguity.
- Decide on an appropriate task order—what makes sense?
- Trial your tasks!
- https://www.userfocus.co.uk/articles/writing-usability-task-scenarios.html

Use task scenarios

- When testing a product, we want to encourage naturalistic behaviour—what we might expect people to do in their own environments.
- Create a realistic scenario that provides a meaningful context for the tasks and can help participants to role play. For example:

  "You don't currently own a car, and you can get almost everywhere you need to go without one.

  You're about to sign up for a class that you can't get to using public transportation, and there are other times when it would be convenient to have a car.

  A friend has been very happy using Zipcar for occasional trips, so you decide to explore that option."

- Then create tasks related to the scenario.

Example scenario source: Steve Krug video: https://www.youtube.com/watch?v=QckIzHC99Xc

How good are these tasks, and why?

If your goal was to find usability problems in the postgraduate courses section of City St George's website, how good would these tasks be?

a) Find a course.

b) Click on the "Prospective Students" option, then "Postgraduate taught degrees" in the drop-down, then "Computing and Computer Science" then browse to find the Human-Computer Interaction Design MSc.

c) You have been working as a designer for 5 years and are now thinking of post-graduate study to gain a UX qualification, possibly part-time. Visit the City St George's website to see if it offers anything of interest.

Task **types** (some)

- Verb-based tasks:
  - Ask the user to do something, and thereby test functionality
  - e.g. *“Today is the deadline for your UCSD coursework. You’ve just finished putting together your report. Submit it to the CW1 area on Moodle”*
- Scavenger-hunt tasks:
  - Ask the user to find something specific
  - Useful for information-rich systems
  - e.g. *“You’ve just moved into a new flat and need to furnish a small living room on a budget. Using the IKEA website, find a sofa you like that seats at least 3 people and costs under £500”*

Adapted from Perfetti (2014)
http://www.slideshare.net/perfettimedia/drupal-con-ux-bootcamp

“Start with the End Goal” (Schade 2017)

*“If you find yourself struggling to write a task, **consider the user’s end goal** rather than the task’s end goal. Rather than focusing on the section or the feature you want to test, consider why people would use that section or feature. What would they ultimately try to accomplish?”*

“To test your checkout process, give the user a task to buy something. To review your newsletter subscription process, ask the user to sign up to receive information via email. To see if a user can understand content, write a task with a question about the information contained in the content. Starting with the users’ end goal helps streamline task writing, and reviewing these common mistakes can finetune the tasks.”

Schade A (2017) https://www.nngroup.com/articles/better-usability-tasks

Task success (& failure) criteria

- Define what "success" means before testing.
  - What is the clear end state of the task?
  - What counts as success vs failure? Is partial success possible?
- Make criteria specific and observable.
  - e.g. correct item selected, task completed, constraints met.
  - Avoid vague goals (e.g. *"find something suitable"*)
  - For some tasks you might even ask participants to tell you when they think/believe a task is complete.
- Align with task design.
  - Success criteria should reflect the scenario and constraints.
  - Ensure tasks are realistic but assessable.
- This enables consistent evaluation across participants, facilitating comparison, and meaningful results.

Planning: Participants

Recruit participants who are real users, or are *representative* of real users.

- Who will you recruit?
- How many do you need?
- How will you recruit them?

See also

- Lecture 3 slides: Participant recruitment
- https://maze.co/blog/user-testing-how-many-users/

Participants: “who should test this?”

- Start with your target users
  - Who is this for? In what context? With what level of experience? ...
  - Do you have existing personas
- Try to recruit people who match key characteristics (role, experience, goals, context...)
- Convenience participants *may* be acceptable in some cases (e.g. for courseworks) if you state the limitations.
- Be honest about representativeness
  - Don’t overclaim: “this proves all users...”
  - Do claim: “this identifies likely usability issues...”

Planning: Ethics

- It's important to treat participants ethically.
  - See lecture 3 slides: Participant recruitment; Ethics & informed consent

- It's important to collect, handle, process, store and destroy data ethically and responsibly.

Ethics: things to consider

- Consent
  - Informed consent, voluntary participation
  - Right to withdraw (and what happens to their data)
- Recording & data handling
  - What you will record (screen, video, audio, notes...)
  - Why you need it (evidence)
  - Where it's stored, who can access it, and how long it's kept.
- Anonymity, including for reporting
  - Remove names, identifying details; use participant codes (P1, P2...)

Planning: Data capture/collection

- Record data:
  - Participant’s interactions with system (computer logging, screen capture, video).
  - Participant’s reactions (video).
  - Thinkaloud or other comments (video / audio).
  - Timings etc.
- In addition, the moderator/facilitator might make notes e.g. just with pen and paper.
- Use questionnaires and interviews for things you cannot observe and things you want to measure.

Planning: Equipment & Materials

Think about the equipment and materials you will need for the sessions, based on what you are planning. This might include:

- Devices & software
  - Test device(s) e.g. laptop, mobile; Prototype or live system; Any required apps / browser
- Recording equipment
  - Screen recording tool; Video camera; Audio recording device
- Session materials
  - Session script; Task scenarios / instructions; Consent form; Participant information sheet; Note-taking template; Pen & paper; Paper prototypes / printouts (if relevant)

Running a usability testing session

1. Use a script.
2. Welcome and brief participant. Check they understand what will happen.
3. Confirm informed consent.
4. Possibly administer pre-test questionnaire.
5. Start recording (e.g. video) at appropriate point—inform participant of this.
6. Ask participants to complete tasks, one at a time, often with a thinkaloud. Brief the tasks verbally, but provide a written copy, as reminder.
7. Observe and record the use of system, remaining neutral.
8. Possibly conduct a debriefing interview (follow-up questions).
9. Administer any post-task/test questionnaire(s).
10. Thank participants for their time.

Generic usability testing script (Steve Krug; mobile version)

Hi, __________. My name is __________, and I'm going to be walking you through this session today.

Before we begin, I have some information for you, and I'm going to read it to make sure that I cover everything.

You probably already have a good idea of why we asked you here, but let me go over it again briefly. We're asking people to try using a mobile app that we're working on so we can see whether it works as intended. The session should take about an hour.

The first thing I want to make clear right away is that we're testing the *app*, not you. You can't do anything wrong here. In fact, this is probably the one place today where you don't have to worry about making mistakes.

As you use the app, I'm going to ask you as much as possible to try to think out loud: to say what you're looking at, what you're trying to do, and what you're thinking. This will be a big help to us.

Also, please don't worry that you're going to hurt our feelings. We're doing this to improve it, so we need to hear your honest reactions.

If you have any questions as we go along, just ask them. I may not be able to answer them right away, since we're interested in how people do when they don't have someone sitting next to them to help. But if you still have any questions when we're done I'll try to answer them then. And if you need to take a break at any point, just let me know.

With your permission, we're going to record what happens on the screen and our conversation. The recording will only be used to help us figure out how to improve the app, and it won't be seen by anyone except the people working on this project. And it helps me, because I don't have to take as many notes.

Also, there are a few people from the design team observing this session in another room. (They can't see us, just the screen.)

If you would, I'm going to ask you to sign a simple permission form for us. It just says that we have your permission to record you, and that the recording will only be seen by the people working on the project.

- ❑ ***Give them a recording permission form and a pen***
- ❑ ***While they sign it, START the SCREEN RECORDER on your laptop***

Do you have any questions so far?

THE QUESTIONS

OK. Before we look at anything, I'd like to ask you just a few quick questions.

- First, what's your occupation? What do you do all day?
- What kind of mobile device (or devices) do you use, like smartphones or a tablet?
- What kinds of things do you spend time doing on your mobile devices?
- Do you have any favorite mobile apps?

THE FIRST SCREEN TOUR

OK, great. We're done with the questions, and we can start looking at things.

First, I'm going to ask you to open up the app labeled ______.

Now, before you start doing anything, just look at the first screen and tell me what you make of it: what strikes you about it, what you think you can do with it, and what it's for. Just look around and do a little narrative.

You can scroll if you want, but please don't "click" (or tap) on anything yet.

- ❑ ***Allow this to continue for two or three minutes, at most.***

THE TASKS

Thanks. Now I'm going to ask you to try doing some specific tasks. I'm going to read each one out loud and give you a printed copy.

I'm also going to ask you to do these tasks without using Search. We'll learn a lot more about how well the site works that way.

And again, as much as possible, it will help us if you can try to think out loud as you go along.

- ❑ ***Hand the participant the first scenario, and read it aloud.***
- ❑ ***Allow the user to proceed until you don't feel like it's producing any value or the user becomes very frustrated.***
- ❑ ***Repeat for each task or until time runs out.***

.....

(see link for rest)

Steve Krug (2010) Rocket Surgery Made Easy
https://sensible.com/download-files/

Usability Testing

Analysing data and reporting

Data analysis: usability problems

- Qualitative data from usability testing include thinkalouds, observations and answers to open questions in interviews and questionnaires:
- Transcribe (or at least review carefully).
- Look for evidence of usability problems in:
  - Observations of interactions with the system (screen capture, video, notes);
  - What participants say, facial expressions and body language;
  - What participants report in interviews and questionnaires.
- Document all usability problems for each participant.
- Group individual problems to remove duplicates, creating a final set.
- Rate the severity of the usability problems.
- Possibly group and categorise problems.

Examples of evidence of usability problems

- The participant does not succeed in a task.
- The participant has to try 2 or more alternative strategies to achieve their goal.
- The participant states what they would do to progress, but it’s not correct.
- The participant takes a less than optimum series of actions.
- The participant expresses surprise.
- The participant expresses frustration etc.
- The participant suggests that something should be changed.

Documenting usability problems

- Create a table / spreadsheet / sticky notes of all usability problems experienced by each participant.
- It should be clear how you found these in the data.
- Be careful to provide a concrete detailed description of every usability problem.
- Describe both the problem *and* its impact on the user.

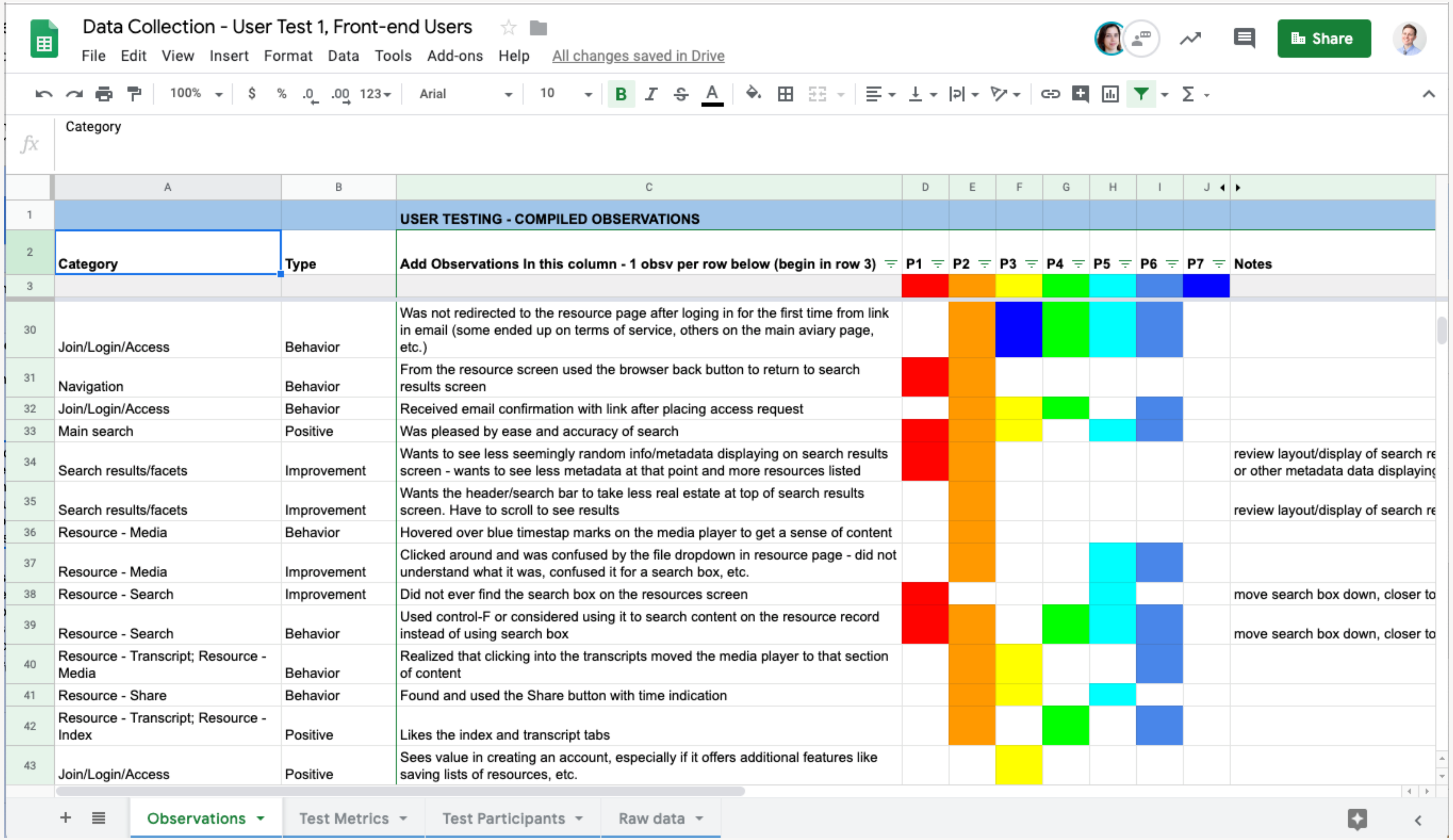
Example: Initial list of usability problems

Ptc	Task	Location	Issue
P4	T1	Journey Planner	Surprised that they could walk to their chosen destination..
P4	T2	Journey Planner	Doesn't feel confident they have the right destination. Says they would use Google maps instead.
P4	T2	Journey Planner	Wanted to select a date in August as task suggested, but couldn't.
P5	T3	Journey Planner	A lot of back and forth searching for relevant ticket information.
P6	T2	Journey Planner	Types in 'Victoria', site interprets this as 'Old Vic'.
P6	T2	Journey Planner	Expects to be able to plan a route outside London.
P7	T2	Journey Planner	Selects the wrong destination but doesn't realise!
P9	T1	Cycling site	Likes the non tourist info i.e. mending your bike.
P10	T3	General	Says they would leave the site to find the zone information.
P10	T4	Interactive Bus Map	Couldn't find the "Print" link on the interactive map (abandons task due to this)
P11	T1	Coach Landing page	Says that links on the page aren't very descriptive.
P11	T1	Cycling site / Bikes and Public Transport	Doubt about the accuracy of information provided (can you take a bike on the train in rush hour).
P11	T2	Journey Planner (on Overground page)	Expected the Journey Planner to only bring back overground jouneys, but others were returned too.
P11	T3	Journey Planner	Expected relevant fares to be be shown in the Journey Planner
P11	T3	Journey Planner	Participant goes to find a zone and then can't find the Tube, DLR, and London Overground Fares again.
P12	T2	Journey Planner	Doesn't know how to amend date. Looks at earlier/later buttons.
P12	T3	Journey Planner	Would like the Journey Planner to help with fares

Group problems to remove duplicates

- Look for duplicate usability problems
- Duplicate problems are when two problems (perhaps experienced by different people) are actually the same problem
- You can use an affinity diagram to help find duplicates
- Create new tables or a rainbow spreadsheet (see next slide) with the duplicate problems removed, but keep track of which participants experienced each problem

Example: Rainbow spreadsheet



	Category	Type	Add Observations In this column - 1 obsv per row below (begin in row 3)	P1	P2	P3	P4	P5	P6	P7	Notes
30	Join/Login/Access	Behavior	Was not redirected to the resource page after loging in for the first time from link in email (some ended up on terms of service, others on the main aviary page, etc.)								
31	Navigation	Behavior	From the resource screen used the browser back button to return to search results screen								
32	Join/Login/Access	Behavior	Received email confirmation with link after placing access request								
33	Main search	Positive	Was pleased by ease and accuracy of search								
34	Search results/facets	Improvement	Wants to see less seemingly random info/metadata displaying on search results screen - wants to see less metadata at that point and more resources listed								review layout/display of search re or other metadata data displaying
35	Search results/facets	Improvement	Wants the header/search bar to take less real estate at top of search results screen. Have to scroll to see results								review layout/display of search re
36	Resource - Media	Behavior	Hovered over blue timestap marks on the media player to get a sense of content								
37	Resource - Media	Improvement	Clicked around and was confused by the file dropdown in resource page - did not understand what it was, confused it for a search box, etc.								
38	Resource - Search	Improvement	Did not ever find the search box on the resources screen								move search box down, closer to
39	Resource - Search	Behavior	Used control-F or considered using it to search content on the resource record instead of using search box								move search box down, closer to
40	Resource - Transcript; Resource - Media	Behavior	Realized that clicking into the transcripts moved the media player to that section of content								
41	Resource - Share	Behavior	Found and used the Share button with time indication								
42	Resource - Transcript; Resource - Index	Positive	Likes the index and transcript tabs								
43	Join/Login/Access	Positive	Sees value in creating an account, especially if it offers additional features like saving lists of resources, etc.								

Image from: https://blog.weareavp.com/handy-tools-for-remote-user-research

How serious are the usability problems?

- Not all usability problems are equal.
- You will probably want to prioritise them:
  - To allocate resources to fixing the most severe problems.
  - To get a feel for the overall usability of the system, and how much redesign work is needed.
- This can be done by rating the severity of each usability problem.

Severity ratings

Severity rating scale (example*):

- Minor: Causes some hesitation or slight irritation.
- Moderate: Causes occasional task failure for some users; causes delays and moderate irritation.
- Critical: Leads to task failure. Causes user extreme irritation.
- Insight/Suggestion/Positive: Users mention an idea or observation that does or could enhance the overall experience.

*From https://measuringu.com/rating-severity/
see this for several alternative (but similar) rating scales

Documenting and reporting usability problems

- Your findings, the usability problems that you identify, will probably have to be understood by someone else. So, they need to be documented clearly.
- What should you include?
  - a description of the problem itself (e.g. cause and outcome).
  - the context in which the problem occurred.
  - the participants who experienced the problem.
  - maybe a screenshot of the issue
  - maybe a rating of its severity.
  - maybe a redesign recommendation.
  - and give it an identifying code or number.

Documenting and reporting usability problems (partial example)

Cause: Lowest price flight is shown for each date, but not the flight times.

Outcome: When comparing prices, the user had to click multiple times to see the price/time combinations for flights on various days (selected date, one day before and one day later) and it was difficult to make a comparison because they were not all visible simultaneously.

Redesign: Replace with a “three day” view – with times and prices for all flights visible.

NO.	PAGE	CAUSE	OUTCOME	REDESIGN	SEVERITY
2.1	Home	Fundraising events is listed in the sub-navigation under 'Support us'	Participants have difficulty finding fundraising events. The first instinct is to scan for events, or get involved.	This requires further testing, perhaps a card sort. Also see 1.1 to make events stand out more from in-page content.	2
2.2	Support us	Searching from the 'support us in your area' module returns all results with sorting.	Participants are overwhelmed by the results and have difficulty finding relevant content, or abandon their search to seek a different method.	Return filtered results instead of sorted results, with the goal of showing fewer more relevant results.	2
2.3	Support us search	Irrelevant results being returned high in the search results list when the user searches using location.	Participants question why they are being showed irrelevant results and waste time double checking to see if they have done things correctly.	Check meta data for events to ensure data is correct. Look into the search feature on the Support us page to ensure it is returning results from relevant channels. Archive old events.	2
2.4	Support us search	Completely irrelevant results being returned when the user searches using keywords and location.	Participant abandoned the journey as 2 results were returned and were of zero relevance to the search queries used.	Check meta data for events to ensure data is correct. Look into the search feature on the Support us page to ensure it is returning results from relevant channels.	1
2.5	Find an event	Only one event type can be selected from the 'Find an event' module.	The participant expressed an interest in being able to choose more than one type of event to improve search efficiency.	Change selector (dropdown) to checklist or similar functionality for multiple selections.	4

Grouping and categorising usability problems

- You may want to group and categorise the usability problems, e.g.
  - all the problems encountered in a specific user journey.
  - or by “type” of issue, e.g. navigation, content, brand, aesthetics, etc.
  - or by area of the system/product: e.g. homepage, checkout, etc.

- Affinity diagrams are a useful method for grouping and categorisation.

(see lecture 4 for more images of usability problem affinity mapping)

Data analysis: Metrics

- Efficiency: Ease of use. e.g. calculated as time for user to perform a task, time to failure or number of interactions
- Effectiveness: How often the user is successful. e.g. calculated as proportion of tasks successfully completed.
- Learnability: Calculated as time required for user to meet a specified level of performance.
- Usability (metrics): e.g. calculated as number of usability problems
- User Experience and Satisfaction: How the user feels about the experience, e.g. calculated from rating scales.

See also https://www.toptal.com/designers/ux/measuring-the-user-experience

Data analysis: Metrics

- Calculate metrics from the quantitative data that you collect:
  - Summarise for each participant
  - Average across participants
  - Report with descriptive statistics
- Report descriptive statistics for small sample sizes appropriately
  - E.g. *‘4/5 participants…’* not *‘80% of participants…’*

What makes a good usability testing report?

- It should state the goals (aims) of the evaluation (i.e. the brief).
- It should explain the method(s) used, and state when the testing was conducted.
- It should describe what was found (evidence relating to the goals).
- Issues should be described clearly, at a useful and usable level of detail.
- Screenshots and participant quotes can be useful.
- The report should provide usability insights, not just recount what happened.
- It should also provide recommendations to address problems identified.
- Language should be clear, straightforward and professional.
- The report should be well-structured, well-styled, and read well.
- Appendices (e.g. detailed table of issues) should be legible and well-labelled.

Questions?

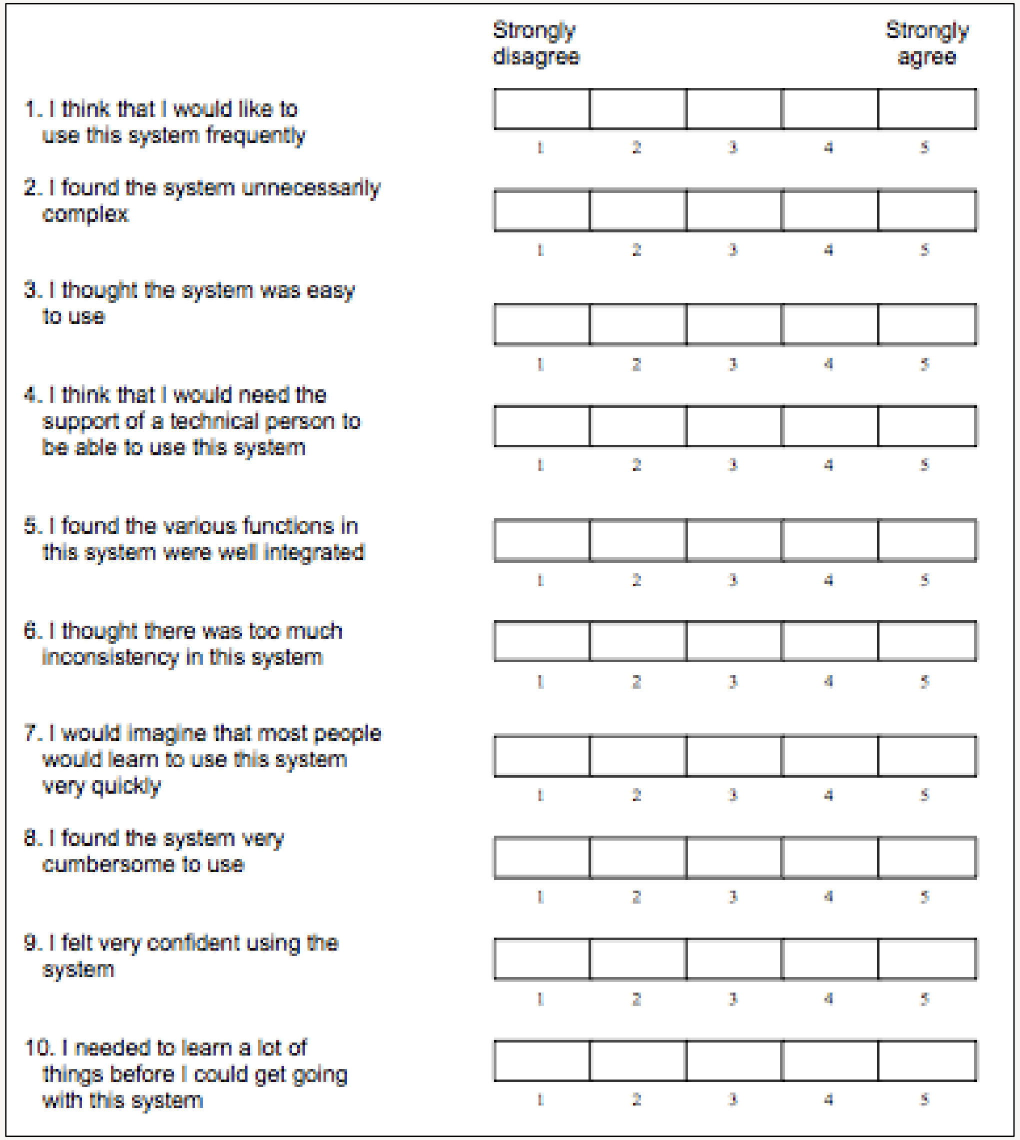
Evaluation:
Usability and UX Questionnaires

Questionnaires in usability & UX evaluation

- We discussed questionnaires as a data collection technique earlier in the module (lecture 3).
- These are often used in usability evaluation
  - To find out about users' perceptions of a system (which may differ from what you observe).
  - To find out about things that you can't observe.
- This is subjective data.
- Often used to supplement other approaches, incl. usability testing.
- You can design questions to focus on specific areas of concern.
- Or you can use a standardised, validated questionnaire, enabling comparison.

System Usability Scale (SUS)

- A classic, still used today.
- Ten items.
- Mixture of positive and negative statements.
- Responses on 5-point Likert scale.
- The aim is to determine an overall score for the system (using an algorithm)
- A score > 68 is indicative of “good” usability.
- Also see https://measuringu.com/interpret-sus-score



	Strongly disagree 1	2	3	4	Strongly agree 5
1. I think that I would like to use this system frequently					
2. I found the system unnecessarily complex					
3. I thought the system was easy to use					
4. I think that I would need the support of a technical person to be able to use this system					
5. I found the various functions in this system were well integrated					
6. I thought there was too much inconsistency in this system					
7. I would imagine that most people would learn to use this system very quickly					
8. I found the system very cumbersome to use					
9. I felt very confident using the system					
10. I needed to learn a lot of things before I could get going with this system					

Single Ease Questionnaire (SEQ)

- Task-level metric: administered immediately after a task
- Rate the difficulty of the task just undertaken
- Short, doesn't disrupt the session flow much.
- Easy to analyse the responses.

From: https://www.nngroup.com/articles/measuring-perceived-usability

User Experience Questionnaire (UEQ)

- Measures user experience.
- Uses a semantic differential scale.
- The image to the right shows the short version (UEQ-S)
- For the full version and more information, see https://www.ueq-online.org

obstructive	○ ○ ○ ○ ○ ○ ○	supportive
complicated	○ ○ ○ ○ ○ ○ ○	easy
inefficient	○ ○ ○ ○ ○ ○ ○	efficient
confusing	○ ○ ○ ○ ○ ○ ○	clear
boring	○ ○ ○ ○ ○ ○ ○	exciting
not interesting	○ ○ ○ ○ ○ ○ ○	interesting
conventional	○ ○ ○ ○ ○ ○ ○	inventive
usual	○ ○ ○ ○ ○ ○ ○	leading edge

Standardised usability/UX questionnaires

There are others; this is just a flavour.

- System Usability Scale (SUS)
- SUMI
- SUPR-Q
- Post-Study System Usability Questionnaire (PSSUQ)
- UEQ
- UMUX and UMUX-Lite
- SEQ
- NASA-TLX
- Net Promoter Score (NPS)

Questions?

Evaluation:
Expert Reviews

Expert reviews

- Not all usability evaluations involve users.
- Expert reviews are evaluations carried out by experts, without users
- They rely on an expert’s judgement
  - But most provide some kind of criteria or framework to support the expert’s judgement.
- Mostly, experts identify the usability problems they *think* will be experienced by the users—predicted usability problems.
- Very limited role in producing usability metrics.
- Sometimes known as “inspection methods”
- Example: Heuristic Evaluation

Heuristic Evaluation

- Influential, expert evaluation technique, developed by Jakob Nielsen
- Originally for conventional ‘desktop’ applications
- Experts inspect the interface to judge its compliance with established usability guidelines (‘heuristics’)
  - E.g. *“Match between system and the real world”*
- Used to predict usability problems—*formative* evaluation.
- Cheaper and faster than usability testing.
- Can unearth serious problems, but also a lot of trivial ones.
- https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation/

Jakob Nielsen’s Heuristics

1. Visibility of System Status
2. Match Between the System and the Real World
3. User Control and Freedom
4. Consistency and Standards
5. Error Prevention
6. Recognition Rather than Recall
7. Flexibility and Efficiency of Use
8. Aesthetic and Minimalist Design
9. Help Users Recognize, Diagnose, and Recover from Errors
10. Help and Documentation

See: https://www.nngroup.com/articles/ten-usability-heuristics.

Preparing for the Heuristic Evaluation

- Create a paper or software prototype
- Recruit several experts (Nielsen suggests 3-5)
- Choose a set of heuristics (different ones are possible)
- Prepare report forms (coding sheets)
- Decide how to do it
  - *Evaluate the whole interface (the original approach)? Or design specific tasks for all evaluators to use? Or allow evaluators to decide on their own tasks?*
- Prepare other materials for use during the evaluation: consent forms, tasks (possibly), instruction sheet, severity rating scale

Conducting the Heuristic Evaluation

- Evaluators inspect interface individually.
- They check whether each screen/feature they encounter violates any of the heuristics.
- Any such violation is considered to be a usability problem.
- They record the problem, its location and the heuristic that was violated: either by writing it down or by describing it to an observer.
- Evaluators might also give a severity rating to each problem.

Dealing with Heuristic Evaluation data

- If you haven't got severity ratings, evaluators should now rate the severity of all problems (not just the ones they found)
- Group similar problems (to reduce the number of problems and to determine how frequently each problem occurs)
- Reassess severity and prioritise
- Recommend possible fixes (redesign)

No	Description	Location	Basis	Severity
1	Inconsistent use of fonts on file and edit menus	Every page	H4	2
2	Can enter non-existent airport into departure field	Book flight page	H5	2

Questions?

Session 9 Summary

- Evaluation is performed to assess the usability and UX of a system, e.g.
  - to identify usability problems that need to be resolved;
  - to obtain quantitative measures of usability and UX (i.e. metrics);
- There are numerous methods that can be used for conducting usability evaluations—some with users, some without users.
- Usability testing is usually the best approach to usability evaluation.
- Usability testing need to be planned and conducted carefully, and the data scrutinised systematically for usability problems.
- Questionnaires and expert reviews also have a role alongside usability testing.